peaches: ChEES-adapted ensemble HMC

A reproducible benchmark of the Hamiltonian walk move vs. `emcee` and NUTS

Abstract

PEACHES is the *Hamiltonian walk move* — an affine-invariant ensemble HMC proposal whose metric is the empirical covariance of the live ensemble — with its integration length tuned by the **ChEES** criterion (Change in the Estimator of the Expected Square; Hoffman, Radul & Sountsov, AISTATS 2021) and its step size by Nesterov dual averaging (Hoffman & Gelman, JMLR 2014). We benchmark it against `emcee` (the gradient-free affine-invariant stretch move; Goodman & Weare 2010, Foreman-Mackey et al. 2013) and NUTS (`numpyro`) on controlled targets with exact moments, over three axes: dimensional scaling, conditioning vs. curvature, and warmup robustness. Every run is checked for convergence (rank-normalized split $\hat{R}$), and every reported efficiency is a median over the *converged* seeds only; all configurations and commands are given for exact reproduction.

1 Methods

peaches: a ChEES-adapted Hamiltonian walk move. Let the ensemble be W walkers $q_{1:W} \in \mathbb{R}^d$ with target $\pi(q) \propto e^{-U(q)}$. The *Hamiltonian walk move* ("h-walk") proposes a Hamiltonian trajectory for a subset of walkers using a momentum that lives in the span of the *complement* ensemble: with C the centered complement walkers (rows $q_j - \bar{q}$), momentum p is a $W \times W$ matrix and the leapfrog updates are

$$q \leftarrow q + \epsilon\, p\, C, \qquad p \leftarrow p - \epsilon\, \nabla U(q)\, C^\top,$$

so positions move only within the affine subspace spanned by the ensemble and the effective metric is the empirical ensemble covariance $\Sigma = \mathrm{cov}(C)$. This makes the move *affine-invariant*: it is unchanged under any invertible linear reparametrization of q, so its efficiency is independent of the target's conditioning. (A complementary "h-side" move proposes along a single random ensemble difference direction; both are Metropolis-corrected with the standard HMC accept ratio.) The preconditioner is recomputed from the live ensemble *every iteration* — there is no separately-fit mass matrix.

Adaptation (warmup only). Three independently-toggleable adapters run during warmup and are then frozen:

1. *Initial step size* via a heuristic line search (`find_init_step_size`): double/halve ϵ until the first trajectory's acceptance brackets the target.

2. *Step size* ϵ via Nesterov dual averaging (Hoffman & Gelman 2014) toward a target acceptance (we pin 0.8 for the scaling law). The log-step update uses gain $\sqrt{t}/\gamma$ on the running acceptance error $\bar{H}$; the realized acceptance is the harmonic mean of per-walker $\min(1, e^{\log \alpha})$.

3. *Integration length* $T = \epsilon L$ via the **ChEES** criterion (Hoffman et al. 2021). ChEES does ADAM ascent on the expected change in squared jump distance: the per-step gradient signal is $g \propto h_t\, T\, (\|c_{\text{pro}}\|^2_\Sigma - \|c_{\text{cur}}\|^2_\Sigma)\, (c_{\text{pro}}^\top \Sigma^{-1} v)$, where c are centered positions, v the trajectory velocity, h_t a Halton sequence, and $\|\cdot\|_\Sigma$ the *affine-invariant* Σ^{-1} norm (matching the move's metric). ADAM (β_1=0, β_2=0.95, lr 0.025, per-step clip $\pm$0.35) updates $\log T$; the realized length is Halton-jittered, $L = \lceil T/\epsilon \rceil$ capped at $L_{\max}$.

Setting all three flags off reduces warmup to a plain burn-in at a fixed (ϵ, L). PEACHES thus realizes *optimally-tuned* ensemble HMC: the metric is free (the ensemble), the step size hits a target acceptance, and the trajectory length maximizes squared jump distance.

Baselines. `emcee` is the gradient-free affine-invariant stretch move (Goodman & Weare 2010; `emcee`, Foreman-Mackey et al. 2013): it is exactly affine-invariant like the h-walk but uses *no* gradient. NUTS is `numpyro`'s implementation with window mass-matrix adaptation; we use the mass matrix appropriate to each experiment (§below) and also an *ablation* with the mass fixed to the identity (`adapt_mass_matrix=False`) to isolate the role of preconditioning.

Targets, metric, convergence. Targets are the isotropic Gaussian $N(0, I_d)$, the rotated ill-conditioned Gaussian (condition number $\kappa = e^{\log \kappa}$), and the Rosenbrock banana — all with exact moments. Efficiency is *production-only* ESS per target-gradient evaluation (ArviZ rank-normalized split ESS, worst coordinate; `emcee` counts one log-density per walker per iteration). Ensemble methods use $n_{\text{walkers}} \propto d$ so the empirical covariance is full rank — exactly $4d$ in the scaling sweep, $\max(128, 4d)$ in the conditioning/warmup experiments (§6) — and a stretch pre-warmup (§5). **Convergence is enforced throughout**: each per-dimension point is the median over its *converged* seeds (rank-normalized split $\hat{R} \leq 1.03$; relaxed slightly from 1.01 for seed-median plus thinning noise, all genuine failures sit at $\hat{R} \geq 1.04$), a stuck seed never enters the median, and a point is drawn solid only if a *majority* of its seeds converged (hollow otherwise — a hollow point's ESS/grad is not a meaningful efficiency).

2 Experiment 1 — dimensional scaling

Setup. Target $N(0, I_d)$ (`--no-rotate --log-cond 0`) for $d \in \{2, 4, \ldots, 256\}$, uniform $n_{\text{walkers}} = 4d$ (no floor). PEACHES pins `--target-accept 0.8` so the step-size law holds at a fixed acceptance; stretch pre-warmup w_s=100, warmup 4000, production 3×10^4 (10^4 at d=256 for device memory; ESS/grad is a ratio, insensitive to production length once converged), $L_{\max} = 512$, 6 seeds at $d \leq 16$ and 3 at $d \geq 32$ (tags `scaleLb`/`scaleL`, `exp4c`). `emcee` mixes far more slowly, so it is run much longer — warmup 2×10^4, production 1×10^5 — and shown only where it reaches $\hat{R} \leq 1.03$ ($d \leq 64$). On the paired Rosenbrock (d=2–32, Fig. 2) PEACHES uses the tuned converged runs (256 walkers for $d \geq 8$, 512 for d=2, 4, warmup $\geq$12000, 6 seeds, tag `scaleLd`); `emcee` reuses tag `exp1`.

Result. Optimally-tuned HMC needs $\epsilon \sim d^{-1/4}$ at fixed acceptance, hence $L \sim d^{1/4}$ and ESS/grad $\sim d^{-1/4}$ (Beskos et al. 2013). PEACHES realizes this in the asymptotic tail: over $d \in [64, 256]$ the fitted efficiency slope is -0.20 (theory -0.25) and $\bar{L}$ grows $7.1 \to 9.4$ ($d^{+0.21}$). Below $d \sim 16$ the curve is pre-asymptotic — with only $4d$ walkers the ensemble covariance is estimated from few samples, so efficiency rises with d to a plateau near $d \sim 32$ before the step-size law takes over. `emcee`'s gradient-free stretch move scales as ESS/grad $\sim d^{-1}$ (fitted -1.26 over the converged $d \in [8, 64]$); already at d=64 PEACHES is $\sim$230$\times$ more efficient per gradient, the gap widens exponentially in d, and `emcee` cannot be converged at all beyond d=64 in a feasible run.

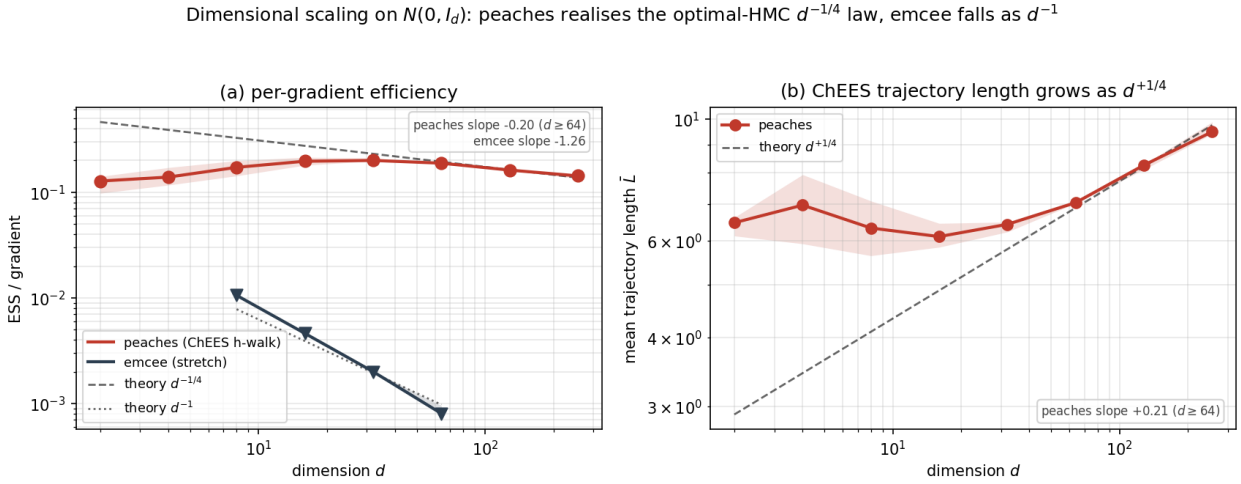


Figure 1: Isotropic Gaussian $N(0, I_d)$, $d = 2$–256 at uniform $n_{\text{walkers}} = 4d$. Median over the converged seeds (6 at $d \leq 16$, 3 at $d \geq 32$); band = seed min–max. **(a)** per-gradient efficiency: PEACHES tracks the optimal-HMC $d^{-1/4}$ law (fitted -0.20 over $d \in [64, 256]$), `emcee` falls as d^{-1} (-1.26) — an exponentially widening gap. The gentle rise for $d \lesssim 16$ is pre-asymptotic: at $4d$ walkers the empirical covariance is estimated from few samples in low dimensions. **(b)** the mechanism: PEACHES's ChEES-tuned mean trajectory length grows as $d^{+1/4}$ (fitted $+0.21$), the signature of the $\epsilon \sim d^{-1/4}$ step-size law. (NUTS also achieves $d^{-1/4}$ and is omitted here so the peaches-vs-emcee contrast reads cleanly; it is benchmarked in §3–4.)

Rosenbrock — the law is masked by curvature. The paired Rosenbrock is formally a product of $d/2$ identical banana pairs, so in principle it obeys the same $d^{-1/4}$ law. It does not over the accessible range $d \leq 32$ (Fig. 2a). With enough walkers (256–512) and a long warmup (≥ 12000) PEACHES converges on *every* seed at $d \geq 8$ ($\hat{R} \leq 1.01$; the single/double-banana cases d=2, 4 reach 5/6), so the efficiency is meaningful — but it stays low ($\sim 10^{-3}$), seed-variable, and drifts *upward* with d rather than falling. The cause (Fig. 2b) is the per-block curvature: it gives ChEES no single good integration length, so the adapted $\bar{L}$ is large and *bimodal* (some seeds find the long trajectory that traverses a banana, $\bar{L} \sim 200$, IAT ~ 1; others a short stuck one, $\bar{L} \sim 20$, IAT ~ 25 — both converged, very different in efficiency), versus the Gaussian's smooth $\bar{L} \sim 6$–9. PEACHES nonetheless stays efficient and converges where `emcee` cannot; the robust curvature statement is the qualitative 2×2 of §3, not a power law.

3 Experiment 2 — conditioning vs. curvature

Setup. (a) Rotated Gaussian d=50, $\kappa = e^{2,4,6,8}$, warmup 1000, max$(128, 4d)$ walkers; PEACHES (production 5000, tag `exp2_lck`), `emcee` (production $6{\times}10^4$), NUTS dense-mass (`exp2_lck`, the standard sampler) and NUTS identity-mass (`--nuts-mass identity`, production 2000, tag `idmass_lck`, the ablation). (b) Rosenbrock $d \in \{8, 16, 32\}$ (tag `exp1`): PEACHES (production 8000, $L_{\max} = 256$ for the long curved trajectories), `emcee` (production 10^5), NUTS identity-mass (`idmass`). On Rosenbrock the global conditioning is mild, so mass adaptation is irrelevant and the identity-mass NUTS represents NUTS faithfully.

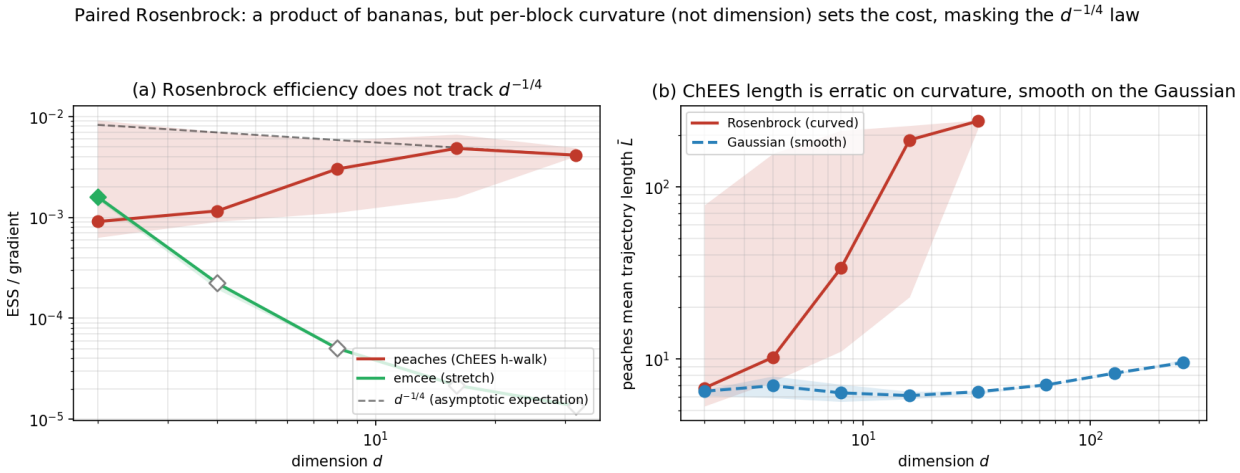


Figure 2: Dimensional scaling on the *paired* Rosenbrock (a product of $d/2$ identical banana pairs). Median over the converged seeds, band = seed min–max; hollow = a minority of seeds converged. **(a)** PEACHES stays efficient (ESS/grad $\sim 10^{-3}$) and converges, but does *not* fall as $d^{-1/4}$ (dashed reference) — it drifts upward; `emcee` fails to converge for $d \geq 4$. **(b)** the mechanism: PEACHES's ChEES length is large and erratic on the curved banana ($\bar{L} \sim$ 10–220, bimodal across seeds) but smooth on the Gaussian ($\bar{L} \sim$ 6–9); the curvature, not the dimension, sets the cost.

Result — a clean 2×2. The two targets isolate two distinct failure modes. *Conditioning* (panel a): the empirical-covariance metric makes PEACHES and `emcee` affine-invariant, so their ESS/grad is flat in κ ($\approx$0.19 and $\approx 10^{-3}$, the latter just dimensionally penalised at d=50). NUTS is flat in κ *only* through its mass adaptation: with the mass fixed to the identity its ESS/grad falls from 3.1×10^{-2} ($\kappa = e^2$) to 3.0×10^{-6} ($\kappa = e^8$) — four orders, and unconverged ($\hat{R} = 1.8$) at $\kappa = e^8$. *Curvature* (panel b): Rosenbrock's position-dependent curvature defeats any fixed linear map. `emcee`'s single affine stretch move cannot cross the ridge — even at 10^5 samples it fails to converge for $d \geq 4$ ($\hat{R}$ grows 1.14 $\to$ 2.0 over d=8 $\to$ 32). PEACHES and NUTS integrate *local* Hamiltonian dynamics along the ridge and stay converged (ESS/grad $\approx 10^{-3}$). Only PEACHES clears both axes: the ensemble preconditioner gives it the conditioning-robustness un-adapted NUTS lacks, and gradient dynamics give it the curvature-handling `emcee` lacks.

4 Experiment 3 — warmup robustness

Setup. Ill-conditioned Gaussian (d=100, $\kappa = e^6$, 400 walkers, tag `exp3`) and Rosenbrock (d=8, 256 walkers, tag `exp3b`); warmup $\in \{50, 100, 200, 500, 1000\}$, production 5000, 8 seeds. NUTS uses a dense mass matrix (the standard choice on an ill-conditioned target). PEACHES uses a 200-step stretch pre-warmup (§5) then its own warmup of the given length.

Result. Because the preconditioner is recomputed from the live ensemble every step, PEACHES needs no mass-matrix warmup: its ESS/grad is flat at $\approx$0.16–0.21 (Gaussian) across all warmup lengths. NUTS-dense must fit a $d(d{+}1)/2$-entry dense mass during warmup; at warmup 50 its ESS/grad is $\approx 2.4 \times 10^{-5}$ — a factor $\sim 10^4$ below PEACHES — and it becomes competitive only once warmup $\geq$500. The same pattern holds on Rosenbrock. This is the practical face of "the metric is free": the work NUTS spends in a long, target-dependent warmup, PEACHES gets for nothing.

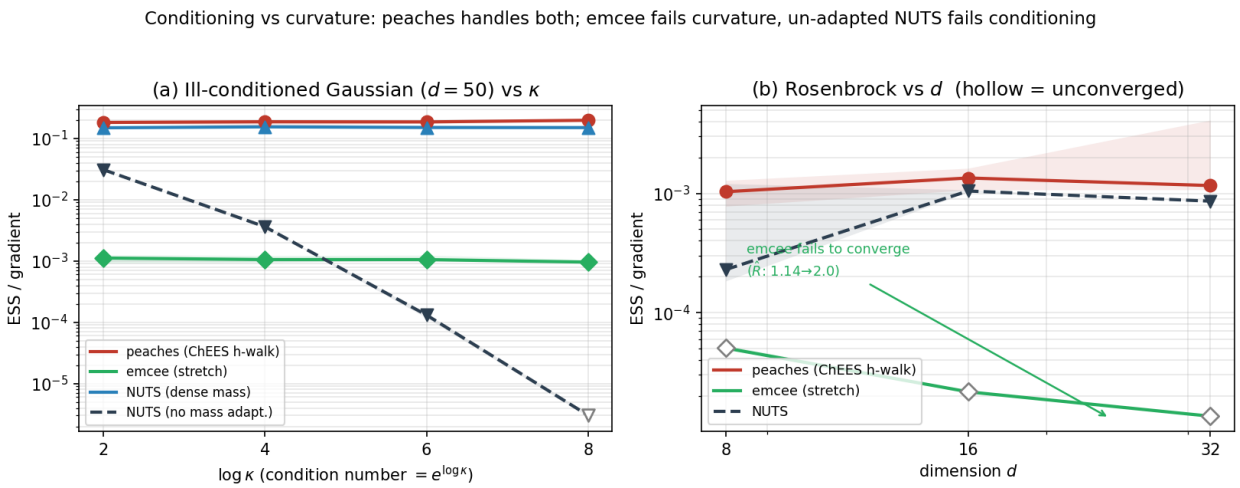


Figure 3: Median over the converged seeds, band = seed min–max; solid = converged ($\hat{R} \leq 1.03$), hollow = not. **(a)** ill-conditioned Gaussian (d=50) vs. κ: PEACHES/`emcee`, and NUTS *with* dense-mass adaptation, are flat in κ; NUTS with the mass fixed to the identity collapses 4 orders. **(b)** Rosenbrock vs. d: PEACHES and NUTS (gradient methods) cross the curved ridge; `emcee` fails to converge for $d \geq 4$.

5 Practical recommendation: a three-phase run

For PEACHES (and ensemble HMC generally) we recommend the following workflow, which the benchmarks above all use:

1. **Stretch pre-warmup (gradient-free).** Run a short `emcee`-style stretch move for w_s iterations *first*, from an over-dispersed initialization. This is cheap (one log-density per walker, no gradient) and affine-invariant, and it does two essential things: it moves the ensemble *mean* onto the typical set (a reasonable centre), and it shapes the ensemble *covariance* to the target's local geometry. This matters because PEACHES's preconditioner *is* the empirical covariance: an isotropic or collapsed initial cloud gives an ill-shaped metric, which in turn collapses the dual-averaged step size and wastes the gradient warmup. A short stretch phase suffices ($w_s \approx$100–200; benefit saturates quickly); it uses no gradient and no knowledge of the target covariance.

2. **peaches warmup (gradient).** With the ensemble now well-centred and well-shaped, run the PEACHES adapters: `find_init_step_size` for the initial ϵ, dual averaging to the target acceptance, and ChEES for the integration length. The empirical-covariance metric is re-estimated each step, so no mass matrix is fit here — only the scalar ϵ and the length L.

3. **Production.** Freeze (ϵ, L) and the adapters and collect samples. ESS/grad is measured here only.

In the code this is exactly (`warm_start` w_s`, warmup` w`, num_samples` N`)`. Phase 1 is the ensemble analogue of NUTS's mass-matrix warmup, but gradient-free and target-agnostic; phase 2 is much shorter and more robust than NUTS's because there is no mass matrix to converge (§4). Skipping phase 1 is the most common failure mode: without it the worst-mixing direction is initialization-dependent and the step size can collapse.

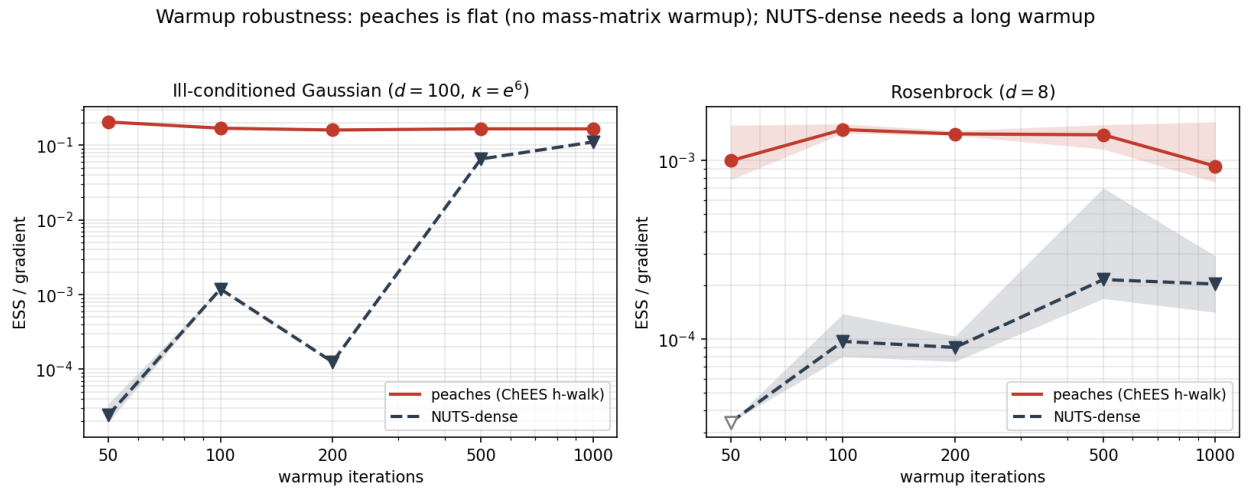


Figure 4: ESS/grad vs. gradient-sampler warmup length, 8 sampling seeds on a fixed target. Line = median over the converged seeds; band = interquartile range (25–75%) over those seeds (the IQR rather than min–max, because ChEES's integration length is bimodal on the banana). Gaussian uses 400 walkers, Rosenbrock 256. PEACHES is flat from warmup 50 to 1000; NUTS-dense needs $\gtrsim$500 warmup steps to fit its mass matrix and is orders less efficient below that (its d=8 Rosenbrock run at warmup 50 barely converges, shown hollow).

6 Reproducibility

All runs go through `python -m benchmark.run` (one `(target,dim,sampler,seed)` cell). Key flags: `--target` {`gaussian,rosenbrock`}, `--dim`, `--sampler` {`peaches,stretch,nuts`}, `--log-cond` ($\log \kappa$; 0 + `--no-rotate` for isotropic), `--n-chains`, `--warm-start` (stretch pre-warmup), `--warmup`, `--num-samples`, `--target-accept`, `--max-L` (PEACHES), `--nuts-mass` {`dense,diag,identity`}, `--max-tree-depth`, `--seed`, `--tag`.

Experiment	target	sampler configs	walkers / warmup
Scaling (§2)	$N(0, I_d)$, d=2–256	PEACHES (w_s=100); `emcee`	$4d$; 4k/30k; emcee
Scaling (§2)	Rosenbrock d=2–32	PEACHES $L_{\max}$=512; `emcee`	256/512; 12k/30k
Conditioning (§3a)	Gauss d=50, κ=$e^{2\text{-}8}$	PEACHES; `emcee`; NUTS `dense`/`identity`	max(128, $4d$); 1k/5
Curvature (§3b)	Rosenbrock $d\in\{8, 16, 32\}$	PEACHES $L_{\max}$=256; `emcee`; NUTS `identity`	max(128, $4d$); 5k/8
Warmup (§4)	Gauss d=100 κ=e^6; Rosen d=8	PEACHES (w_s=200); NUTS `dense`	400/256; {50..1000

Scaling uses 6 seeds at $d \leq 16$ (3 elsewhere); the warmup experiment uses 8 seeds; the rest use 3. Per-dimension efficiency is the median over the converged seeds ($\hat{R} \leq 1.03$); the warmup figure's band is the interquartile range, all others are seed min–max. Convergence ($\hat{R}$), bias, and std-error are stored alongside ESS/grad in each `metrics.json`. Figures are produced by `benchmark/plot_pn_*.py`.

References. Beskos, Pillai, Roberts, Sanz-Serna, Stuart (2013), *Optimal tuning of the hybrid Monte Carlo algorithm*, Bernoulli 19(5A). Goodman & Weare (2010), *Ensemble samplers with affine invariance*, Comm. App. Math. Comp. Sci. 5(1). Foreman-Mackey, Hogg, Lang, Goodman (2013), *emcee: the MCMC hammer*, PASP 125. Hoffman & Gelman (2014), *The No-U-Turn Sampler*, JMLR 15. Hoffman, Radul, Sountsov (2021), *An adaptive-MCMC scheme for setting trajectory lengths in Hamiltonian Monte Carlo*, AISTATS, PMLR 130.